Step	Lever (task)	Measured result (exploratory)	Verdict
I-0	frozen encoder, linear head	dev_x AUC 0.693 video, 0.610 speech	baseline
I-1	full fine-tuning (video); I-1b, I-1c	dev_x 0.910, gain +0.170; I-1b 0.932 and I-1c 0.919 do not enter	supported
I-2	trim and normalise (speech)	dev_x 0.672, gain +0.062	supported
I-3	noise, codec, reverb copies (speech)	dev_x 0.746, gain +0.073	supported
I-4	average of three encoders (speech)	dev_x 0.813, gain +0.068	supported
I-5	frozen heads on LAV-DF, then fusion	video head 0.384, below chance	not tested
I-6	heads trained on LAV-DF, fusion	visual head at chance within produced videos	see text
I-7	rPPG signal check, positive control	0.301 of real clips, 0.6 needed; control sensitive	failed
I-8	integration and freeze	sealed test: T1, T2 improved; T3 superior at 95 percent	supported

the lever of each step was chosen from the diagnostic battery of the step before; verdicts are mechanical and exploratory